

Bayesian Optimization on Multi-Fidelity Data

Making every precise measurement more valuable

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Begin: A Story

Your job is to tune some parameters, but:

- Only about fifteen expensive experiments allowed
- Tune many interacting parameters — more than ten
- Real in engines, materials, ...
- Question: spend a tiny budget in the smartest way?

Background 1: Black-Box and Budget

Why is it HARD?

- **Black box:** you see input and output, never the inside
- **Budget:** only fifteen precise runs, but a huge space
- Task: find the best point of a high-dimensional black box

Background 2: Two Channels

Good news: Two Channels:

$$h_k(\mathbf{x}) = h_{k-1}(\mathbf{x}) + \delta_k(\mathbf{x})$$

- Cheap channel: run many times, but biased
- Expensive channel: only about 15 runs, accurate
- Idea: combine them — high = low + bias

The Skeleton Comes from One Person



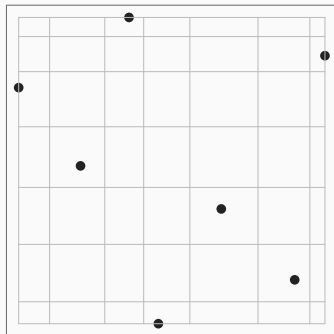
- V. R. Joseph, Georgia Tech
- Maxpro design — where to start
- Sequential design — add points one by one
- KOH fusion — join cheap and expensive data

Main Tools 1: Maxpro Design

The criterion of maxpro design:

$$\min_D \sum_{i=1}^n \prod_{j \neq i} \frac{1}{\sum_{k=1}^p (x_{ik} - x_{jk})^2}$$

- Start evenly: spread in every projection



Main Tools 2: Gaussian Process

Definition

Gaussian Process A random function $Y(\mathbf{x})$ is a **Gaussian process** if for any choice of $\mathbf{x}_1, \dots, \mathbf{x}_n$ and any $n \geq 1$, $\mathbf{Y} = (Y(\mathbf{x}_1), \dots, Y(\mathbf{x}_n))'$ has a multivariate normal distribution.

- Surrogate: Gaussian process = a normal distribution over functions
- It predicts a value, AND an uncertainty, at every point

The Whole Process

The whole pipeline:

1. Spread out the starting points
2. Feed in many cheap runs $\rightarrow$ a rough map
3. Send one expensive run to the predicted valley
4. Repeat — the map gets sharper every round

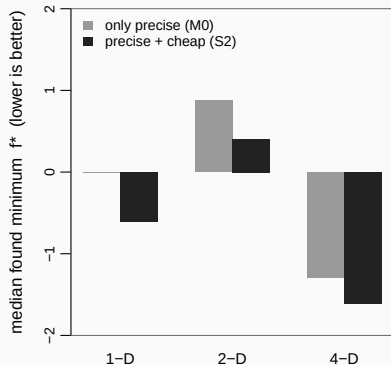
Our Question

Our Question:

$$\hat{\mathbf{x}} \approx \arg \min_{\mathbf{x} \in [0,1]^p} h_{\text{H}}(\mathbf{x})$$

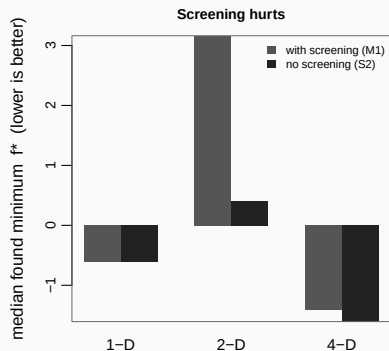
- Goal: minimize the expensive function, with only 15 precise runs
- Three small questions:
 - Does combining the two channels help?
 - Does screening the parameters help?
 - Does it still work in high dimensions?

Answer 1: Combination Is the Key



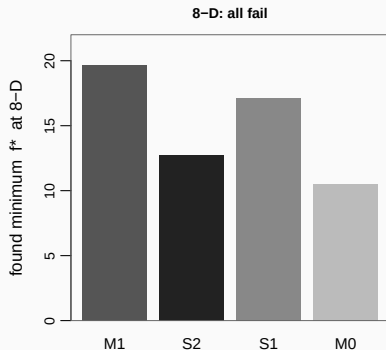
- Metric: median of the best found, over 20 repeats
- Cheap + precise beats precise alone
- 1-D: precise alone stuck at 0; with fusion $\rightarrow$ global optimum
- 2-D: lands right next to the known answer

Answer 2: Screening Is Useless



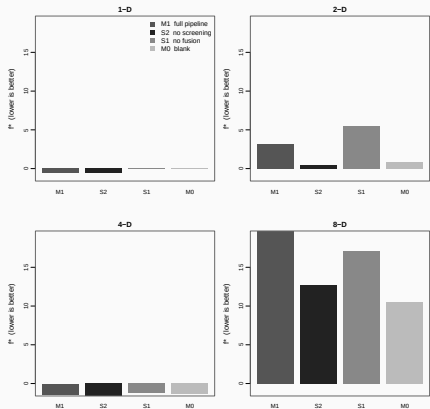
- Screening ranks parameters by cheap runs
- But cheap runs are biased $\rightarrow$ the ranking is unreliable
- Result: removing screening actually helps

Answer 3: High Dimensions Fail



- In 8-D, fusion collapses too
- Too few precise points to sketch an 8-D landscape
- Fusion is decisive — but it has a ceiling

Global Picture



- Low dims: full pipeline wins; drop fusion $\rightarrow$ breaks
- High dims: nothing works
- One line: make every precise test count; limit = dimensionality

My Future Directions:

1. When is screening trustworthy? (find the bias conditions)
2. Tame high dimensions? (add structure, or reduce dimension first)
3. Real-world tool: give it a budget — get the best plan

Summary

- Not “more precise runs” — but “more value per run”
- Limit: the curse of high dimensions

Thank you!